

PRISMA 2020 Checklist

Manuscript: Matrix stiffness responses localize to human osteoblast lineage programs without a universal gene signature

Adapted for a revision-stage systematic search and secondary quantitative synthesis of public functional-genomics datasets. Completed 9 September 2026. PRISMA 2020 source: Page et al., BMJ 372, n71 (2021); official checklist at <https://www.prisma-statement.org/prisma-2020-checklist> (CC BY 4.0).

item	topic	PRISMA requirement	location	status/comment
1	Title	Identify the report as a systematic review.	Title and Abstract	Partial: title states the biological synthesis; systematic identification is explicit in Abstract/Methods.
2	Abstract	Provide a structured summary.	Abstract	Objectives, eligibility scale, methods, main estimates and limitations reported in compact journal format.
3	Rationale	Describe rationale in context.	Introduction	Reported.
4	Objectives	State explicit objectives.	Introduction, final paragraph	Reported.
5	Eligibility criteria	Specify inclusion/exclusion criteria and grouping.	Methods: Protocol and study eligibility	Reported.
6	Information sources	Specify databases and date last searched.	Methods: Systematic search and screening	GEO and PubMed; searched 9 September 2026.
7	Search strategy	Present full strategies.	Methods; archive search_audit/search_methods_and_flow.md	Exact strings reported.
8	Selection process	Describe screening process.	Methods; Supplementary Fig. 1; screening log	Single-reviewer computational project; every decision and reason retained.
9	Data collection process	Describe data extraction.	Methods: Expression processing	Scripts and accession-level provenance supplied.
10	Data items	List outcomes and other variables.	Methods; Supplementary Tables 1 and 14	Reported.
11	Risk of bias	Describe assessment of study limitations.	Supplementary Table 14; Discussion	Reporting assessment used; no validated tool exists for this mechanobiology omics design.
12	Effect measures	Specify effect measures.	Methods: Expression processing	Hedges-corrected standardized mean differences; paired standardized differences.
13	Synthesis methods	Describe eligibility, preparation, displays, synthesis and heterogeneity.	Methods: Gene and pathway meta-analysis	REML, modified Hartung-Knapp, study clustering and sensitivity analyses reported.
14	Reporting bias assessment	Describe methods.	Discussion	Formal funnel tests were not used with seven heterogeneous studies; stated as a limitation.
15	Certainty assessment	Describe certainty methods.	Discussion; Interpretive safeguards	Triangulation, exact inference and sensitivity concordance used; no GRADE rating applied.
16	Study selection	Report search and selection results.	Supplementary Fig. 1; screening log	46 screened, 12 full assessed, 7 included.
17	Study characteristics	Cite and describe included studies.	Supplementary Table 1; Data availability	Reported by accession and design.
18	Risk of bias in studies	Present assessments.	Supplementary Table 14	Reported as public-record reporting assessment.
19	Results of individual studies	Present summary statistics/effect estimates.	Figure 2c; Supplementary Tables 6, 11 and machine tables	Reported.
20	Results of syntheses	Summarize contributors, heterogeneity and sensitivity.	Results; Figures 1-3; Supplementary Tables	Reported with exact P values and sensitivity analyses.
21	Reporting biases	Present assessment.	Discussion	Not estimable reliably at k=7; limitation stated.
22	Certainty of evidence	Present certainty.	Discussion	Claims calibrated as supported, partial, borderline or unsupported.
23	Discussion	Interpret results, limitations and implications.	Discussion	Reported.
24	Registration and protocol	Provide registration/protocol information.	Methods; protocol archive	Not registered; revision-stage search stated; dated analysis protocols supplied.
25	Support	Describe financial/nonfinancial support.	Acknowledgements	Author completion required.
26	Competing interests	Declare competing interests.	Competing interests	Author completion required.
27	Availability	Report public availability of materials.	Data and Code availability; accompanying archive	Accessions and archived release supplied; version DOI 10.5281/zenodo.22726636.

Items marked for author completion are visible in red placeholders in the manuscript or cover letter and must be resolved before submission.